Enter to Learn, Go out to Serve

NOTIFICATION of ASSESSMENT

Course	**Software Engineering**		
Task name	Progressive Web App		
Task Type	Hand in & In class	**Issue Date**	**Wednesday, 11 March**
Task Number	2	**Date Due**	**Friday, 24 April**

Components and Weighting

	Component	**Weighting**
1	PWA Hand in	10
2	Programming for the Web Quiz	10

Task Description

Component 1: PWA Hand in

Learn from the mistakes of others. You can't live long enough to make them all yourself.

- Eleanor Roosevelt

With the countless data breaches that have occurred in recent years, it is worth learning from and examining each in turn. In order to provide a resource for Software Engineering students, a cybersecurity dashboard PWA has been commissioned. Here are the requirements:

- The dashboard should display at least 3 of the OWASP Top 10 web application vulnerabilities (vulnerabilities page)
- When the category is clicked on, it should take the user to another page that lists data breaches as a result of the chosen vulnerability (Incidents page)
- Each incident should have a link that takes the user to the article about that breach
- The user should be able to add an incident from the incident page
- A way on the incidents page to show trend analysis on the data (e.g. how many incidents in a particular year)

AI use

AI use is permitted but only through the use of the provided Gem in Gemini and for the code building component only. These must be cited. You must not use AI content in your responses to the questions in the report.

Link to the Gem: https://gemini.google.com/gem/1dE9fte0zTIBJDF_Evsm-EL1zkxL5w_Hf?usp=sharing

What you need to submit:

- ☐ Completed Cyberwatch Report
- ☐ Trello board with comment logs. Make a copy of this one:
  - ☐ https://trello.com/invite/b/69aba7c014afe3c800cc4e25/ATTI8373fa1b468ac195e3b503a4fb3400304052C46A/12se-pwa-task
- ☐ Github repository link to your PWA with regular, meaningful commit messages. Ensure your repository was cloned from the original.

Component 2: Quiz

There will be an in class quiz on the due date. The topic that will be assessed is the content that has been covered from the Programming For the Web unit.

Submission Details

Reduced marks or zero may be awarded for late submission as per the Assessment policy.

Outcomes to be assessed

SE-12-01, SE-12-02, SE-12-03, SE-12-04, SE-12-05, SE-12-06, SE-12-07, SE-12-08, SE-12-09

Ai Statement

Component 1

3	AI ASSISTED COMPLETION	AI may be used to help complete the task. Students must actively review, adapt, and improve AI outputs to show their own understanding. **You must critically evaluate and modify any AI-generated content you use. Any AI created content must be cited.**

Component 2

1	NO AI	The assessment is completed entirely without AI assistance in a controlled environment, ensuring that students rely solely on their existing knowledge, understanding, and skills **You must not use AI at any point during the assessment. You must demonstrate your core skills and knowledge.**

Marking Rubric

Justifies workflow adjustments based on the movement and priority of task cards in the logbook.	Refines the interaction between the Flask backend and frontend to ensure seamless data flow (e.g. no crashes when invalid data is entered)		Evaluates how the chosen framework and UI features enhance the accessibility and inclusivity of the app.	Evaluates the success of accessibility interventions by comparing pre- and post-implementatio n scores.	Synthesises final audit data to explain how specific PWA technologies influenced the app's performance.		*Students at this level apply sophisticated technical skills in the design & implementation of programming along with comprehensive skills in developing, managing and documenting software engineering projects*
Synchronises commit messages with Trello task cards to document specific technical changes.	Constructs a filtering system to update the dashboard based on user input.	Implements GROUP BY and/or aggregate functions to synthesise raw data into dashboard metrics.	Explains the selection of CSS components by linking them to a specific UI/UX design principle	Documents the specific code changes and technical steps taken to resolve accessibility warnings.	Uses Lighthouse to confirm the app is "Installable" and functions correctly in an "Offline" state.	Justifies the selection of tools (Flask, SQL, Bootstrap) based on project requirements.	*Students at this level apply high-level technical skills in the design & implementation of programming solutions and demonstrate high-level skills in developing, managing and documenting software engineering projects*
Updates Trello cards to reflect the progression of features from "In Progress" to "Done."	Implements a functional input form that inserts a new incident record into the database.	Constructs SQL queries using WHERE constraints and JOIN to link multiple data tables.	Implements UI features, such as a grid layout, using specific CSS framework components.	Describes the impact of identified accessibility issues on the end-user's experience.	Documents the configuration of manifest.json and sw.js to enable basic PWA functionality.	Explains how specific sections of code contribute to the application's overall functionality.	*Students at this level apply sound technical skills in the design & implementation of programming solutions and demonstrates sound skills in developing, managing and documenting software engineering projects*
Records task completion status within a Trello board and GitHub commit history.	Builds a Flask interface using Jinja and Python renders HTML pages and lists vulnerability data from the database.	Writes SQL SELECT statements to retrieve and display raw data from a single database table.	Identifies specific sections of HTML code where a CSS framework (e.g., Bootstrap) has been applied.	Records initial Lighthouse scores and identifies the accessibility issues highlighted in the report.	Identifies the core requirements for a Progressive Web App, including manifest and service workers.	Describes the basic purpose and primary functions of the developed web application.	*Students at this level display basic skills in programming and basic skills in developing software engineering projects*
Insufficient evidence	Insufficient evidence	Insufficient evidence	Insufficient evidence	Insufficient evidence	Insufficient evidence	Insufficient evidence	
Project Management & Version Control	Functional Web Application & Logic	Data Engineering & SQL Execution	Technical Design & Frameworks	Accessibility & Initial Audit Analysis	PWA Optimisation & Implementation	Interview	Level statements

SE-12-01, SE-12-09	SE-12-02, SE-12-07	SE-12-04, SE-12-07	SE-12-08	SE-12-05, SE-12-06	SE-12-08, SE-12-03	SE-12-05, SE-12-02	